*For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory*

# Irinotecan Hydrochloride Injection

# IRINOTEL

**SAFETY WARNINGS: Diarrhea and Myelosuppression**

Early and late forms of diarrhea can occur. Early diarrhea may be accompanied by cholinergic symptoms which may be prevented or ameliorated by atropine. Late diarrhea can be life threatening and should be treated promptly with loperamide. Monitor patients with diarrhea and give fluid and electrolytes as needed. Institute antibiotic therapy if patients develop ileus, fever, or severe neutropenia. Interrupt irinotecan and reduce subsequent doses if severe diarrhea occurs.

Severe myelosuppression may occur.

## DESCRIPTION:

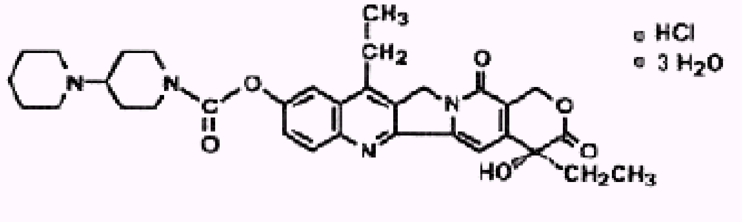
Irinotecan is a semisynthetic derivative of the plant alkaloid camptothecin extracted from the Chinese tree *Camptotheca accuminata*. Irinotecan is a promising antitumor agent with activity in a broad range of experimental tumor models. It inhibits topoisomerase I function by binding to the topoisomerase I / DNA-cleavable complex. Irinotecan Hydrochloride Injection is a light yellow colored solution free from visible particles.

## COMPOSITION:

Each ml contains:

| | |
|---|---|
| Irinotecan Hydrochloride Trihydrate | 20.00mg |
| Sorbitol USNF | 45.00mg |
| Lactic Acid USP | 0.9mg |
| Water for Injection USP | q.s. to 1.0 ml |

## CHEMICAL STRUCTURE:



Irinotecan Hydrochloride

Irinotecan hydrochloride is a pale yellow-to-yellow crystalline powder. Chemically it is (***S***)-4,11-diethyl-3,4,12,14-tetrahydro-4-hydroxy-3,14-dioxo1*H* pyrano[3',4':6,7]-indolizino[1,2-b]quinolin-9-yl-[1,4'bipiperidine]-1'-carboxylate, monohydrochloride, trihydrate. It is slightly soluble in water and organic solvents. It has an empirical formula of $C_{33}H_{38}N_4O_6.HCl.3H_2O$ and a molecular weight of 677.19.

## PHARMACOLOGY:

**Mechanism of Action:**
Irinotecan is a derivative of camptothecin. Camptothecins interact specifically with the enzyme topoisomerase I which relieves torsional strain in DNA by inducing reversible single-strand breaks. Irinotecan and its active metabolite SN-38 bind to the topoisomerase I-DNA complex and prevent religation of these single-strand breaks. Current research suggests that the cytotoxicity of irinotecan is due to double-strand DNA damage produced during DNA synthesis when replication enzymes interact with the ternary complex formed by topoisomerase I, DNA, and either irinotecan or SN-38. Mammalian cells cannot efficiently repair these double-strand breaks.

Irinotecan serves as a water-soluble precursor of the lipophilic metabolite SN-38. SN-38 is formed from irinotecan by carboxylesterase-mediated cleavage of the carbamate bond between the camptothecin moiety and the dipiperidino side chain. SN-38 is approximately 1000 times as potent as irinotecan as an inhibitor of topoisomerase I purified from human and rodent tumor cell lines. *In vitro* cytotoxicity assays show that the potency of SN-38 relative to irinotecan varies from 2- to 2000-fold. However, the plasma area under the concentration versus time curve (AUC) values for SN-38 are 2% to 8% of irinotecan and SN-38 is 95% bound to plasma proteins compared to approximately 50% bound to plasma proteins for irinotecan. The precise contribution of SN-38 to the activity of irinotecan is thus unknown. Both irinotecan and SN-38 exist in an active lactone form and an inactive hydroxy acid anion form. A pH-dependent equilibrium exists between the two forms such that an acid pH promotes the formation of the lactone, while a more basic pH favors the hydroxy acid anion form. Administration of irinotecan has resulted in antitumor activity in mice bearing cancers of rodent origin and in human carcinoma xenografts of various histological types.

**Pharmacokinetics:**

After intravenous infusion of irinotecan in humans, irinotecan plasma concentrations decline in a multiexponential manner, with a mean terminal elimination half-life of about 6 to 12 hours. The mean terminal elimination half-life of the active metabolite SN-38 is about 10 to 20 hours. The half-lives of the lactone (active) forms of irinotecan and SN-38 are similar to those of total irinotecan and SN-38, as the lactone and hydroxy acid forms are in equilibrium.

Over the recommended dose range of 50 to 350 $mg/m^2$, the AUC of irinotecan increases linearly with dose; the AUC of SN-38 increases less than proportionally with dose. Maximum concentrations of the active metabolite SN-38 are generally seen within 1 hour following the end of a 90-minute infusion of irinotecan.

***Distribution***
Irinotecan exhibits moderate plasma protein binding (30% to 68% bound). SN-38 is highly bound to human plasma proteins (approximately 95% bound). The plasma protein to which irinotecan and SN-38 predominantly binds is albumin.

***Metabolism***
Irinotecan is subject to extensive metabolic conversion by various enzyme systems, including esterases to form the active metabolite SN-38, and UGT1A1 which mediates the glucuronidation of SN-38 to form an inactive metabolite. SN-38 glucuronide had 1/50 to 1/100 the activity of SN-38. Patients who are homozygous for either the UGT1A1*28 or *6 alleles, or who are compound heterozygous for these alleles, have higher SN-38 AUC than patients with the wild-type UGT1A1 alleles. Irinotecan can also undergo CYP3A4-mediated oxidative metabolism to several inactive

metabolites, one of which can be hydrolyzed by carboxylesterase to release the active metabolite SN-38.

***Excretion***
The disposition of irinotecan has not been fully elucidated in humans. The urinary excretion of irinotecan is 11% to 20%; SN-38, <1%; and SN-38 glucuronide, 3%. The cumulative biliary and urinary excretion of irinotecan and its metabolites (SN-38 and SN-38 glucuronide) over a period of 48 hours following administration of irinotecan in two patients ranged from approximately 25% (100 $mg/m^2$) to 50% (300 $mg/m^2$).

**Pharmacokinetics in Special Populations**

***Geriatric Patients***
No change in the starting dose is recommended for geriatric patients receiving the weekly dosage schedule of irinotecan.

***Male and Female Patients***
The pharmacokinetics of irinotecan do not appear to be influenced by gender.

***Racial and Ethnic Groups***
The influence of race on the pharmacokinetics of irinotecan has not been evaluated.

***Patients with Hepatic Impairment***
Irinotecan clearance is diminished in patients with hepatic dysfunction while exposure to the active metabolite SN-38 is increased relative to that in patients with normal hepatic function. The magnitude of these effects is proportional to the degree of liver impairment as measured by elevations in total bilirubin and transaminase concentrations. However, the tolerability of irinotecan in patients with hepatic dysfunction (bilirubin greater than 2 mg/dl) has not been assessed sufficiently.

***Patients with Renal Impairment***
The influence of renal insufficiency on the pharmacokinetics of irinotecan has not been evaluated.

**Pharmacogenomics**
The active metabolite SN-38 is further metabolized via UGT1A1. Genetic variants of the UGT1A1 gene such as the UGT1A1*28 [(TA)7] and *6 alleles lead to reduced UGT1A1 enzyme expression or activity and decreased function to a similar extent.

Individuals who are homozygous or compound (double) heterozygous for these alleles (e.g., *28/*28, *6/*6, *6/*28) are UGT1A1 poor metabolizers and are at increased risk for severe or life-threatening neutropenia from irinotecan due to elevated systemic exposure to SN-38.

The UGT1A1*6/*6 genotype should not be confused with 6/6 genotype, which is sometimes used to represent the genotype of individuals who are wild type for UGT1A1*28. Individuals who are heterozygous for either the UGT1A1*28 or *6 alleles (*1/*6, *1/*28) are UGT1A1 intermediate metabolizers and may also have an increased risk of severe or life-threatening neutropenia. Individuals with UGT1A1*28 and *6 alleles may be at an increased risk of severe diarrhea. The risk evidence appears greater in UGT1A1*28 and *6 homozygous patients and in those taking irinotecan doses > 125 $mg/m^2$.

UGT1A1*28 and *6 alleles occur at various frequencies in different populations. Approximately 20% of Black or African American, 10% of White, and 2% of East Asian individuals are homozygous for the UGT1A1*28 allele. Approximately 2-6 % of East Asian individuals are homozygous for the UGT1A1*6 allele. The UGT1A1*6 allele is uncommon in Black or African American or in White individuals. Decreased function alleles other than UGT1A1*28 and *6 may be present in certain populations.

**INDICATIONS:**
IRINOTEL (Irinotecan Injection) is indicated for the treatment of patients with advanced colorectal cancer:
• In combination with 5-fluorouracil and folinic acid in patients without prior chemotherapy for advanced disease.
• As a single agent in patients who have failed an established 5-fluorouracil containing treatment regimen.

**CONTRAINDICATIONS:**
IRINOTEL (Irinotecan Injection) is contraindicated in patients with a known hypersensitivity to the drug or its excipients.

**ADVERSE EFFECTS:**

**Clinical Studies Experience**
Common adverse reactions observed in combination therapy are nausea, vomiting, abdominal pain, diarrhea, constipation, anorexia, mucositis, neutropenia, leukopenia (including lymphocytopenia), anemia, thrombocytopenia, asthenia, pain, fever, infection, abnormal bilirubin, and alopecia.

Common adverse reactions observed in single agent therapy are nausea, vomiting, abdominal pain, diarrhea, constipation, anorexia, neutropenia, leukopenia (including lymphocytopenia), anemia, asthenia, fever, body weight decreasing, and alopecia.

The most clinically significant adverse events for patients receiving irinotecan-based therapy were diarrhea, nausea, vomiting, neutropenia, and alopecia. The most clinically significant adverse events for patients receiving 5-fluorouracil (5-FU)/ folinic acid (FA) therapy were diarrhea, neutropenia, neutropenic fever, and mucositis. Grade 4 neutropenia, neutropenic fever (defined as grade 2 fever and grade 4 neutropenia), and mucositis were observed less often with weekly irinotecan/5-FU/FA than with monthly administration of 5-FU/FA.

**Other adverse reactions**
The following adverse reactions have been identified during post approval use of Irinotecan. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

Myocardial ischemic events have been observed following irinotecan therapy. Thromboembolic events have been observed in patients receiving irinotecan.

Symptomatic pancreatitis, asymptomatic pancreatic enzyme elevation have been reported. Increases in serum levels of transaminases (i.e., AST and ALT) in the absence of progressive liver metastasis have been observed.

Hyponatremia, mostly with diarrhea and vomiting, has been reported.

Transient dysarthria has been reported in patients treated with Irinotecan; in some cases, the event was attributed to the cholinergic syndrome observed during or shortly after infusion of irinotecan.

Interaction between Irinotecan and neuromuscular blocking agents cannot be ruled out. Irinotecan has anticholinesterase activity, which may prolong the neuromuscular blocking effects of suxamethonium and the neuromuscular blockade of non-depolarizing drugs may be antagonized.

Infections: fungal and viral infections have been reported.

**DRUG INTERACTIONS:**

***5-fluorouracil (5-FU) and leucovorin (LV):***
The disposition of irinotecan was not substantially altered when the drugs were co-administered. Although the $C_{max}$ and $AUC_{0-24}$ of SN-38, the active metabolite, were reduced when irinotecan was followed by 5-FU and FA administration compared with when irinotecan was given alone, this sequence of administration is recommended.

**Strong CYP3A4 Inducers**
Exposure to irinotecan or its active metabolite SN-38 is substantially reduced in adult and pediatric patients concomitantly receiving the CYP3A4 enzyme-inducing anticonvulsants phenytoin, phenobarbital, carbamazepine, or St. John’s wort. The appropriate starting dose for patients taking these or other strong inducers such as rifampin and rifabutin has not been defined. Consider substituting non-enzyme inducing therapies at least 2 weeks prior to initiation of irinotecan therapy. Do not administer strong CYP3A4 inducers with irinotecan unless there are no therapeutic alternatives.

**Strong CYP3A4 or UGT1A1 Inhibitors**
Irinotecan and its active metabolite, SN-38, are metabolized via the human cytochrome P450 3A4 isoenzyme (CYP3A4) and uridine diphosphate-glucuronosyl transferase 1A1 (UGT1A1), respectively. Patients receiving concomitant ketoconazole, a CYP3A4 and UGT1A1 inhibitor, have increased exposure to irinotecan and its active metabolite SN-38. Coadministration of irinotecan with other inhibitors of CYP3A4 (e.g., clarithromycin, indinavir, itraconazole, lopinavir, nefazodone, nelfinavir, ritonavir, saquinavir, telaprevir, voriconazole) or UGT1A1 (e.g., atazanavir, gemfibrozil, indinavir) may increase systemic exposure to irinotecan or SN-38. Discontinue strong CYP3A4 inhibitors at least 1 week prior to starting irinotecan therapy. Do not administer strong CYP3A4 or UGT1A1 inhibitors with irinotecan unless there are no therapeutic alternatives.

**PRECAUTIONS AND WARNINGS:**

**Diarrhea and Cholinergic Reactions**
Early diarrhea (occurring during or shortly after infusion of irinotecan) is usually transient and infrequently severe. It may be accompanied by cholinergic symptoms of rhinitis, increased salivation, miosis, lacrimation, diaphoresis, flushing, and intestinal hyperperistalsis that can cause abdominal cramping. Bradycardia may also occur. Early diarrhea and other cholinergic symptoms may be prevented or treated. Consider prophylactic or therapeutic administration of 0.25 mg to 1 mg of intravenous or subcutaneous atropine (unless clinically contraindicated). These symptoms are expected to occur more frequently with higher irinotecan doses.

Late diarrhea (generally occurring more than 24 hours after administration of irinotecan) can be life threatening since it may be prolonged and may lead to dehydration, electrolyte imbalance, or sepsis. Grade 3-4 late diarrhea occurred in 23-31 % of patients receiving weekly dosing. In the clinical studies, the median time to the onset of late diarrhea was 5 days with 3-week dosing and 11 days with weekly dosing. Late diarrhea can be complicated by colitis, ulceration, bleeding, ileus, obstruction, and infection. Cases of megacolon and intestinal perforation have been reported. Patients should have loperamide readily available to begin treatment for late diarrhea. Begin loperamide at the first episode of poorly formed or loose stools or the earliest onset of bowel movements more frequent than normal. One dosage regimen for loperamide is 4 mg at the first onset of late diarrhea and then 2 mg every 2 hours until the patient is diarrhea-free for at least 12 hours. Loperamide is not recommended to be used for more than 48 consecutive hours at these doses, because of the risk of paralytic ileus. During the night, the patient may take 4 mg of loperamide every 4 hours. Monitor and replace fluid and electrolytes. Use antibiotic support for ileus, fever, or severe neutropenia. Subsequent weekly chemotherapy treatments should be delayed in patients until return of pretreatment bowel function for at least 24 hours without anti-diarrhea medication. Patients must not be treated with irinotecan until resolution of the bowel obstruction. If grade 2, 3, or 4 late diarrhea recurs, subsequent doses of irinotecan should be decreased. Avoid diuretics or laxatives in patients with diarrhea.

**Myelosuppression**

Irinotecan can cause severe myelosuppression. Bacterial, viral, and fungal infections have occurred in patients treated with irinotecan.

Deaths due to sepsis following severe neutropenia have been reported in patients treated with irinotecan. In the clinical studies evaluating the weekly dosage schedule, neutropenic fever (concurrent NCI grade 4 neutropenia and fever of grade 2 or greater) occurred in 3 % of the patients; 6 % of patients received G-CSF for the treatment of neutropenia. Manage febrile neutropenia promptly with antibiotic support. Hold irinotecan if neutropenic fever occurs or if the absolute neutrophil count drops $<1000/mm^3$. After recovery to an absolute neutrophil count $\geq 1000/mm^3$, subsequent doses of irinotecan should be reduced.

Based on sparse available data, the concurrent administration of irinotecan with irradiation is not recommended.

Patients with baseline serum total bilirubin levels of 1.0 mg/dL or more also had a greater likelihood of experiencing first-cycle grade 3 or 4 neutropenia than those with bilirubin levels that were less than 1.0 mg/dL (50 % versus 18 %; $p<0.001$). Patients with deficient glucuronidation of bilirubin, such as those with Gilbert’s syndrome, may be at greater risk of myelosuppression when receiving therapy with irinotecan.

**Increased Risk of Neutropenia in Patients with Reduced UGT1A1 Activity**

Individuals who are homozygous for either the UGT1A1*28 or *6 alleles (*28/*28, *6/*6) or who are compound or double heterozygous for the UGT1A1*28 and *6 alleles (*6/*28) are at increased risk for severe or life-threatening neutropenia during treatment with irinotecan. These individuals are UGT1A1 poor metabolizers and experience increased systemic exposure to SN-38, an active metabolite of irinotecan. Individuals who are heterozygous for either the UGT1A1*28 or *6 alleles (*1/*28, *1/*6) are intermediate metabolizers and may also have an increased risk of severe or life-threatening neutropenia.

Consider UGT1A1 genotype testing for the *28 and *6 alleles to determine UGT1A1 metabolizer status.

When administering irinotecan, consider a reduction in the irinotecan starting dose by at least one level for patients known to be homozygous or compound heterozygous for the UGT1A1*28 and/or *6 alleles (*28/*28, *6/*6, *6/*28).

Closely monitor patients with UGT1A1*28 or *6 alleles for neutropenia during and after treatment with irinotecan. The precise dosage reduction in this patient population is not known. Subsequent dosage modifications may be required based on individual patient tolerance to treatment.

**Hypersensitivity**
Hypersensitivity reactions including severe anaphylactic or anaphylactoid reactions have been observed. Discontinue irinotecan if anaphylactic reaction occurs.

**Renal Impairment/Renal Failure**
Rare cases of renal impairment and acute renal failure have been identified, usually in patients who became volume depleted from severe vomiting and/or diarrhea.

**Pulmonary Toxicity**
Interstitial Pulmonary Disease (IPD)-like events, including fatalities, have been reported in patients receiving irinotecan (in combination and as monotherapy) for treatment of colorectal cancer and other advanced solid tumors. In the event of an acute onset of new or progressive, unexplained pulmonary symptoms such as dyspnea, cough, and fever, irinotecan and other co-prescribed chemotherapeutic agents should be interrupted pending diagnostic evaluation. If IPD is diagnosed, irinotecan and other chemotherapy should be discontinued and appropriate treatment instituted as needed.

**Toxicity of the 5 Day Regimen**
Irinotecan Injection should not be used in combination with a regimen of 5-FU/FA administered for 4-5 consecutive days every 4 weeks because of reports of increased toxicity, including toxic deaths.

**Increased Toxicity in Patients with Performance Status 2**
In patients receiving either irinotecan/5-FU/FA or 5-FU/FA, higher rates of hospitalization, neutropenic fever, thromboembolism, first-cycle treatment discontinuation, and early deaths were observed in patients with a baseline performance status of 2 than in patients with a baseline performance status of 0 or 1.

**Embryofetal Toxicity**
Based on its mechanism of action and findings in animals, irinotecan can cause fetal harm when administered to a pregnant woman. In animal studies, intravenous administration of irinotecan during the period of organogenesis resulted in embryofetal mortality and teratogenicity in pregnant animals at exposures lower than the human exposure based on area under the curve (AUC) at the clinical dose of 125 mg/m$^2$. Advise pregnant women of the potential risk to a fetus.

Advise female patients of reproductive potential to avoid becoming pregnant and to use highly effective contraception during treatment with irinotecan and for 6 months after the final dose. Advise male patients with female partners of reproductive potential to use condoms during treatment and for 3 months after the final dose of irinotecan.

**Patients with Hepatic Impairment**
The use of irinotecan in patients with significant hepatic impairment has not been established. Irinotecan was not administered to patients with serum bilirubin >2.0 mg/dL, or transaminase >3 times the upper limit of normal if no liver metastasis, or transaminase >5 times the upper limit of normal with liver metastasis.

**Carcinogenesis, Mutagenesis and Impairment of Fertility**

Long-term carcinogenicity studies with irinotecan were not conducted. Rats were, however, administered intravenous doses of 2 mg/kg or 25 mg/kg irinotecan once per week for 13 weeks (in separate studies, the 25 mg/kg dose produced an irinotecan $C_{max}$ and AUC that were about 7.0 times and 1.3 times the respective values in patients administered 125 mg/m$^2$ weekly) and were then allowed to recover for 91 weeks.

No significant adverse effects on fertility and general reproductive performance were observed after intravenous administration of irinotecan in doses of up to 6 mg/kg/day to rats and rabbits.

**Pregnancy**
Risk Summary
Based on findings and its mechanism of action irinotecan can cause fetal harm when administered to a pregnant woman. Available postmarketing and published data reporting the use of irinotecan in pregnant women, are insufficient and confounded by the concomitant use of other cytotoxic drugs, to evaluate for any drug-associated risk for major birth defects, miscarriage, or adverse maternal or fetal outcomes.

**Females and Males of Reproductive Potential**

<u>Pregnancy Testing</u>
Verify the pregnancy status in female patients of reproductive potential prior to initiating irinotecan.

<u>Contraception</u>
Irinotecan can cause fetal harm when administered to a pregnant woman.

Females
Advise female patients of reproductive potential to use effective contraception during treatment and for 6 months after the final dose of irinotecan.

Males
Due to the potential for genotoxicity, advise male patients with female partners of reproductive potential to use condoms during treatment and for 3 months after the final dose of irinotecan.

<u>Infertility</u>

Females
Based on postmarketing reports, female fertility may be impaired by treatment with irinotecan. Menstrual dysfunction has been reported following irinotecan administration.

Males
Based on findings from animal studies, male fertility may be impaired by treatment with irinotecan.

**Lactation**
Risk Summary
Irinotecan and its metabolites are present in human milk. There is no information regarding the effects of irinotecan on the breastfed infant, or on milk production. Because of the potential for serious adverse reactions from irinotecan in the breastfed child, advise lactating women not to breastfeed during treatment with irinotecan and for 7 days after the final dose.

**Pediatric Use**
The effectiveness of irinotecan in pediatric patients has not been established.

**Geriatric Use**
Patients greater than 65 years of age should be closely monitored because of a greater risk of late diarrhea in this population. The starting dose of irinotecan in patients 70 years and older for the once-every-3-week-dosage schedule should be 300 mg/m$^2$.

**Renal Impairment**
The influence of renal impairment on the pharmacokinetics of irinotecan has not been evaluated. Therefore, use caution in patients with impaired renal function. Irinotecan is not recommended for use in patients on dialysis.

**Hepatic Impairment**
Irinotecan clearance is diminished in patients with hepatic impairment while exposure to the active metabolite SN-38 is increased relative to that in patients with normal hepatic function. The magnitude of these effects is proportional to the degree of liver impairment as measured by elevations in total bilirubin and transaminase concentrations. Therefore, use caution when administering irinotecan to patients with hepatic impairment. The tolerability of irinotecan in patients with hepatic dysfunction (bilirubin greater than 2 mg/dl) has not been assessed sufficiently, and no recommendations for dosing can be made.

## DOSAGE AND ADMINISTRATION:

Strictly follow the recommended dosage unless directed otherwise by the physician. All doses of irinotecan should be administered as an intravenous infusion over 30 to 90 minutes.

Single-agent dosage schedules
Single-agent dosage schedules have been extensively studied for metastatic colorectal cancer.

**Starting dose**
*Weekly Dosage Schedule.* The recommended single-agent starting dose of irinotecan is 125 mg/m$^2$. A lower starting dose may be considered (e.g., 100 mg/m$^2$) for patients with any of the following conditions: prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Treatment should be given in repeated 6-week cycles, comprising weekly treatment for 4 weeks, followed by a 2-week rest.

*Once-Every-2-Week Dosage Schedule.* The usual recommended starting dose of irinotecan is 250 mg/m$^2$ every 2 weeks by intravenous infusion. A lower starting dose may be considered (e.g., 200 mg/m$^2$) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer.

*Once-Every-3-Week Dosage Schedule*. The usual recommended starting dose of irinotecan for the onceevery-3-week dosage schedule is 350 mg/m$^2$. A lower starting dose may be considered (e.g., 300 mg/m$^2$) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer.

**Special populations Elderly**

The dose should be chosen carefully in this population due to their greater frequency of decreased biological functions. This population should require more intensive surveillance (See section Pharmacokinetic Properties).

**Patients with impaired hepatic function**

In patients with hepatic dysfunction, the following starting doses are recommended:

**Table 1: Starting Doses in Patients with Hepatic Dysfunction: Single-agent Weekly Regimen**

| **Serum Total Bilirubin Concentration** | **Serum ALT/AST Concentration** | **Starting Dose, mg/m$^2$** |
|---|---|---|
| 1.5-3.0 x IULN | ≤5.0 x IULN | 60 |
| 3.1-5.0 x IULN | ≤5.0 x IULN | 50 |
| <1.5 x IULN | 5.1-20.0 x IULN | 60 |
| 1.5-5.0 x IULN | 5.1-20.0 x IULN | 40 |

**Table 2: Starting Doses in Patients with Hepatic Dysfunction: Single-agent Once-Every-3 Week Regimen**

| **Serum Total Bilirubin Concentration** | **Starting Dose, mg/m$^2$** |
|---|---|
| 1.5-3.0 x IULN | 200 |
| >3.0 x IULN | Not Recommended[a] |

[a] The safety and pharmacokinetics of irinotecan given once-every-3-weeks have not been defined in patients with bilirubin >3.0 x institutional upper limit of normal (IULN) and this schedule cannot be recommended in these patients.

**Patients with impaired renal function**

Studies in this population have not been conducted (See section Pharmacokinetic Properties). Therefore, caution should be undertaken in patients with impaired renal function. Irinotecan is not recommended for use in patients on dialysis.

**Combination-agent dosage schedules Starting Dose**

*Irinotecan in Combination with 5-Fluorouracil (5-FU) and folinic acid (FA).* Irinotecan in combination with 5-FU and folinic acid is recommended for use in patients with metastatic colorectal cancer. For all regimens, the dose of folinic acid should be administered immediately after irinotecan, with the administration of 5-FU to occur immediately after receipt of folinic acid. The currently recommended regimens are shown below:

*Regimen 1 (6-week cycle with bolus 5-FU/FA):* The recommended starting dose is 125 $mg/m^2$ of irinotecan, 500 $mg/m^2$ bolus 5-FU, and 20 $mg/m^2$ bolus folinic acid.

*Regimen 2 (6-week cycle with infusional 5-FU/FA):* The recommended starting dose is 180 $mg/m^2$ of irinotecan, 400 $mg/m^2$ bolus 5-FU, 600 $mg/m^2$ 5-FU infusion, and 200 $mg/m^2$ folinic acid.

Lower starting doses may be considered for irinotecan (e.g., 100 $mg/m^2$) and 5-FU (e.g., 400 $mg/m^2$) for patients with any of the following conditions: age 65 years and older, prior extensive radiotherapy, performance status of 2, increased bilirubin levels, or gastric cancer. Treatment should be given in repeated 6-week cycles, comprising weekly treatment for 4 weeks, followed by a 2-week rest.

**Duration of treatment**

For both single-agent and combination-agent regimens, treatment with additional cycles of irinotecan may be continued indefinitely in patients who attain a tumor response or in patients whose cancer remains stable. Patients should be carefully monitored for toxicity and should be removed from therapy if unacceptable toxicity occurs that is not responsive to dose modification and routine supportive care.

**Dose modification recommendations**

The recommended dose modifications during a cycle of therapy and at the start of each subsequent cycle of therapy for single-agent dosage schedules are described in **Table 3**. These recommendations are based on toxicities commonly observed with the administration of irinotecan. For modifications at the start of a subsequent cycle of therapy, the dose of irinotecan should be decreased relative to the initial dose of the previous cycle.

The recommended dose modifications during a cycle of therapy and at the start of each subsequent cycle of therapy for irinotecan, 5-FU, and folinic acid are described in **Table 4**.

All dose modifications should be based on the worst preceding toxicity. A new cycle of therapy should not begin until the toxicity has recovered to Grade 2 or less. Treatment may be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicity. If the patient has not recovered, consideration should be given to discontinuing irinotecan.

**Table 3: Recommended Dose Modifications for Single-agent Schedules**

A new cycle of therapy should not begin until the granulocyte count has recovered to $\geq 1500/mm^3$, and the platelet count has recovered to $\geq 100{,}000/mm^3$, and treatment-related diarrhea is fully resolved. Treatment should be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicities. If the patient has not recovered after a 2-week delay, consideration should be given to discontinuing irinotecan.

| Toxicity<br>NCI Grade[b] (Value) | During a Cycle of Therapy | At the Start of the Next Cycle of Therapy (After Adequate Recovery), Compared with the Starting Dose in the Previous Cycle[a] | |
|---|---|---|---|
| | **Weekly** | **Weekly** | **Once Every 2 or 3 Week** |
| No toxicity | Maintain dose level | ↑ 25 mg/m² up to a maximum dose of 150 mg/m² | Maintain dose level |
| Neutropenia<br>1 (1500 to 1999/mm³)<br>2 (1000 to 1499/mm³)<br>3 (500 to 999/mm³)<br>4 (<500/mm³) | <br>Maintain dose level<br>↓ 25 mg/m²<br>Omit dose, then ↓ 25 mg/m² when resolved to ≤Grade 2<br>Omit dose, then ↓ 50 mg/m² when resolved to ≤Grade 2 | <br>Maintain dose level<br>Maintain dose level<br>↓ 25 mg/m²<br>↓ 50 mg/m² | <br>Maintain dose level<br>Maintain dose level<br>↓ 50 mg/m²<br>↓ 50 mg/m² |
| Neutropenic fever (Grade 4 neutropenia & | Omit dose, then ↓ 50 mg/m² when resolved | ↓ 50 mg/m² | ↓ 50 mg/m² |
| ≥Grade 2 fever) | | | |
| Other hematologic toxicities | Dose modifications for leukopenia, thrombocytopenia, and anemia during a cycle of therapy and at the start of subsequent cycles of therapy are also based on NCI toxicity criteria and are the same as recommended for neutropenia above. | | |
| Diarrhea<br>1 (2-3 stools/day > pretx[c])<br>2 (4-6 stools/day > pretx[c])<br>3 (7-9 stools/day > pretx[c])<br>4 (≥ 10 stools/day > pretx[c]) | <br>Maintain dose level<br>↓ 25 mg/m²<br>Omit dose, then ↓ 25 mg/m² when resolved to ≤Grade 2<br>Omit dose, then ↓ 50 mg/m² when resolved to ≤Grade 2 | <br>Maintain dose level<br>Maintain dose level<br>↓ 25 mg/m²<br>↓ 50 mg/m² | <br>Maintain dose level<br>Maintain dose level<br>↓ 50 mg/m²<br>↓ 50 mg/m² |
| Other non-hematologic toxicities[d]<br>1<br>2<br>3<br>4 | <br>Maintain dose level<br>↓ 25 mg/m²<br>Omit dose, then ↓ 25 mg/m² when resolved to ≤Grade 2<br>Omit dose, then ↓ 50 mg/m² when resolved to ≤Grade 2 | <br>Maintain dose level<br>↓ 25 mg/m²<br>↓ 25 mg/m²<br>↓ 50 mg/m² | <br>Maintain dose level<br>↓ 50 mg/m²<br>↓ 50 mg/m²<br>↓ 50 mg/m² |

[a] All dose modifications should be based on the worst preceding toxicity.

[b] National Cancer Institute Common Toxicity Criteria.

[c] Pre-treatment.

[d] Excludes alopecia, anorexia, asthenia.

**Table 4: Recommended Dose Modifications for Irinotecan/5-Fluorouracil/Folinic Acid Combination Schedules**

Patients should return to pre-treatment bowel function without requiring antidiarrhea medications for at least 24 hours before the next chemotherapy administration. A new cycle of therapy should not begin until the granulocyte count has recovered to ≥1500/mm³, and the platelet count has recovered to ≥100,000/mm³, and treatment-related diarrhea is fully resolved. Treatment should be delayed 1 to 2 weeks to allow for recovery from treatment-related toxicities. If the patient has not recovered after a 2-week delay, consideration should be given to discontinuing irinotecan.

| Toxicity NCI Grade[b] (Value) | During a Cycle of Therapy | At the Start of Subsequent Cycles of Therapy |
|---|---|---|
| No toxicity | Maintain dose level | Maintain dose level |
| Neutropenia<br>1 (1500 to 1999/mm³)<br>2 (1000 to 1499/mm³)<br>3 (500 to 999/mm³)<br>4 (<500/mm³)<br><br>Neutropenic fever (Grade 4 neutropenia & ≥Grade 2 fever) | <br>Maintain dose level[c]<br>↓ 1 dose level[d]<br>Omit dose, then ↓ 1 dose level when resolved to ≤Grade 2<br>Omit dose, then ↓ 2 dose levels when resolved to ≤Grade 2[d]<br><br>Omit dose, then ↓ 2 dose levels when resolved | <br>Maintain dose level[c]<br>Maintain dose level<br>↓ 1 dose level[d]<br>↓ 2 dose levels<br><br>↓ 2 dose levels |
| Other hematologic toxicities | Dose modifications for leukopenia or thrombocytopenia during a cycle of therapy and at the start of subsequent cycles of therapy are also based on NCI toxicity criteria and are the same as recommended for neutropenia above. | |
| Diarrhea<br>1 (2-3 stools/day >pretx[e])<br>2 (4-6 stools/day >pretx)<br>3 (7-9 stools/day >pretx)<br>4 (≥10 stools/day >pretx) | <br>Delay dose until resolved to baseline (bsl), then give same dose<br>Omit dose, then ↓ 1 dose level when resolved to bsl<br>Omit dose, then ↓ 1 dose level when resolved to bsl<br>Omit dose, then ↓ 2 dose levels when resolved to bsl | <br>Maintain dose level<br>Maintain dose level<br>↓ 1 dose level<br>↓ 2 dose levels |
| Other non-hematologic Toxicities[f] | | |
| 1<br>2<br>3<br>4 | Maintain dose level<br>Omit dose, then ↓ 1 dose level when resolved to ≤Grade 1<br>Omit dose, then ↓ 1 dose level when resolved to ≤Grade 2<br>Omit dose, then ↓ 2 dose levels when resolved to ≤Grade 2<br><br>*For mucositis/stomatitis decrease only 5-FU, not irinotecan[g]* | Maintain dose level<br>Maintain dose level<br>↓ 1 dose level<br>↓ 2 dose levels<br><br>*For mucositis/ stomatitis decrease only 5-FU, not irinotecan[g]* |

[a]Dose modification refers to irinotecan and 5-FU; LV dose remains fixed at 20 mg/m² (not adjusted).

[b]National Cancer Institute Common Toxicity Criteria.

[c]Refers to initial dose used in previous cycle.

[d]Irinotecan: dose level reductions = 25 mg/m² decrements; 5-Fluorouracil: dose level reductions = 100 mg/m² decrements.

[e]Pre-treatment.

[f]Excludes alopecia, anorexia, asthenia.

[g]For mucositis/stomatitis decrease only 5-FU, not irinotecan

**Dosage in Patients with Reduced UGT1A1 Activity.**

When administered in combination with other agents, or as a single-agent, consider a reduction in the starting dose by at least one level of Irinotecan Hydrochloride Injection, USP for patients known to be homozygous for the UGT1A1*28 or *6 alleles (*28/*28, *6/*6) or compound heterozygous for the UGT1A1*28 and *6 alleles (*6/*28) (see sections **Posology and method of administration**, **Special warnings and precautions for use**, and **Pharmacological Properties)**. Subsequent dosage modifications may be required based on individual patient tolerance to treatment **(see section** See section **Dosage and Administration**).

**Preparation of Infusion Solution**

Inspect vial contents for particulate matter and discoloration and repeat inspection when drug product is withdrawn from vial into syringe. Irinotecan Injection 20 mg/ml is intended for single use only and any unused portion should be discarded.

Irinotecan Injection must be diluted prior to infusion using aseptic technique. Irinotecan Injection should be diluted in 5 % Dextrose Injection, (preferred) or 0.9 % Sodium Chloride Injection, to a final concentration range of 0.12 mg/ml to 3.0 mg/ml. Other drugs should not be added to the infusion solution.

Prepare the infusion solution immediately prior to use and commence infusion as soon as possible after preparation. If visible particulates are present in the infusion solution discard. If it is not possible to use the infusion solution immediately, the infusion solution may be stored for up to 24 hours at 2 °C to 8 °C or discarded.

**OVERDOSE:**
Single doses of up to 750 mg/m$^2$ of irinotecan were administered to patients with various cancers. The adverse events in these patients were similar to those reported with the recommended dosage and regimen. There have been reports of overdosage at doses up to approximately twice the recommended therapeutic dose, which may be fatal. The most significant adverse reactions reported were severe neutropenia and severe diarrhea. There is no known antidote for overdosage of irinotecan. Maximum supportive care should be instituted to prevent dehydration due to diarrhea and to treat any infectious complications.

**STORAGE:**
Store below 30ºC. Protect from light. Do not freeze. The vials should remain in the carton until the time of use. Physical and chemical stability is maintained for 24 hours at room temperature and after dilution. Refrigeration (2 to 8$^o$ C) and protection from light of solutions prepared in 5 % Dextrose Injection, or 0.9 % Sodium Chloride Injection confers stability for 48 hours. Use IRINOTEL prepared solutions within 24 hours if stored at room temperature. Avoid freezing of IRINOTEL and administered solutions as it may result in precipitation of the drug. Physical admixture with other drugs is not recommended. Visually inspect for particulate matter, precipitation, or discoloration prior to use.

**INCOMPATIBILITIES:**
This medicinal product must not be mixed with other medicinal products except those mentioned in this package insert.

**SHELF LIFE:**
The shelf life of unopened vials is 2 years.

**After dilution:**
Chemical and physical in-use stability has been demonstrated for 24 hours below 25°C and for 48 hours at 2°C to 8°C.
From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C to 8°C, unless the dilution has taken place in controlled and validated aseptic conditions.

**HANDLING AND DISPOSAL:**
Irinotecan is a potent chemotherapeutic anti cancer agent. All standard procedures applicable for proper handling of anticancer agents should be considered.

- Irinotecan should be handled only by persons conversant with standard practices of handling and disposal of anticancer agents. The use of gloves is recommended.

- In case, irinotecan comes in contact with skin or mucosa, wash the skin or mucosa thoroughly with soap and water.

**PRESENTATION:**
IRINOTEL, a sterile, light yellow clear aqueous solution, is available as 2-ml and 5-ml injection containing 40 mg and 100 mg of Irinotecan HCl trihydrate respectively at a concentration of 20 mg/ml.

**REFERENCES:**
US Prescribing information of Camptosar (Irinotecan) Injection, Pharmacia & Upjhon Co., Division of Pfizer Inc, NY, 10017, 01/2022.

Summary of Product Characteristics, CAMPTO 40 mg/2 mL, 100 mg/5 mL and ~~300~~ mg/15 mL Concentrate for solution for infusion, Pfizer (Perth) Pty Limited, Australia, 13 JUL 2022.

**MANUFACTURER INFORMATION:**

**Manufactured in India by:**
**Fresenius Kabi Oncology Limited**
Village-Kishanpura
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